Supplementary Material: Communication as the Binding Constraint

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I. S1. Relation to the superseded version

An earlier manuscript ("Nested Federated Learning" / "TierFed" v2) reported results produced by code with the following defects, found in an internal audit and corrected here: (1) differential privacy was applied inside two strategies only and clipped the aggregate rather than each client's update, so FedAvg/FedPer/FedProx ran without noise and the reported budgets were invalid; the TierFed DP runs crashed on local-only rounds and the table was unsupported; (2) the SCAFFOLD baseline was FedAvg with doubled bit accounting; (3) the scalability table contained no TierFed runs and its values were not those of the logs; (4) the architecture sweep compared different depths across methods; (5) the imaging result used an undisclosed configuration on a binarised 97%-positive task; (6) the inversion attack targeted the fixed schedule at initialisation and was labelled TierFed; (7) the model described (93 inputs) was not the model trained (162 inputs); (8) repeated runs were not deterministic; (9) no patient-level deduplication was performed and one "missing specialty" client held 49% of the data; (10) the greedy controller reached only powers of two; (11) the fixed multi-rate schedule sent a fallback tier whenever no tier was due, which made its medium-tier period ineffective; (12) compressors were applied to raw weights instead of updates; (13) in v3, the DP comparison ran data-dependent schedulers (adaptive TierFed, AdaComm) under the Gaussian accountant, which does not charge for the information their schedules reveal, so their reported budgets were not valid DP guarantees; v4 refuses such strategies under DP, adds the data-independent TierFed-DP planner, and replaces the equal-σ table by a matched-ε comparison with Poisson sampling (Section S3). Items (1)–(10) were found in an audit of the earlier manuscript, items (11)–(12) during the v3 runs and item (13) in the v4 revision; all affected runs were deleted and rerun. With these corrections the earlier central finding, that fixed multi-rate schedules lose about 0.05 AUROC to personalised baselines, does not reproduce and is retracted. Every item maps to a code change in the released repository (`ROADMAP.md`, section 0) and to a test.

II. S2. Derivation of Proposition 1

We follow the cooperative-SGD framework of Wang and Joshi [JMLR 2021, Thm. 1], which bounds the averaged gradient norm of local-update SGD in terms of the per-round deviation of client iterates from the virtual average. Write the shared parameter as $\theta = (\theta_g)_g$ with disjoint blocks. For site k and block g let $\theta_{k,g}^{(t)}$ be the iterate after t local steps since block g was last synchronised and $\bar{\theta}_g^{(t)}$ the (virtual) average across sites.

Lemma 1 (Blockwise deviation)**.** *Under A1–A3, for* $\eta \le c_0/(LK_{\max}\tau_{\max})$ *and any* $t \le K_g\tau_{\max}$,

$$\frac{1}{N}\sum_k \mathbb{E}\big\|\theta_{k,g}^{(t)} - \bar{\theta}_g^{(t)}\big\|^2 \;\le\; c\,\eta^2 t^2\,(\Gamma_g + \sigma_g^2) \;\le\; c\,\eta^2 K_g^2 \tau_{\max}^2 (\Gamma_g + \sigma_g^2).$$

Proof. Identical to Lemma 2 of Wang and Joshi restricted to the coordinates of block g: the deviation after t steps is a sum of t stochastic-gradient differences, each with second moment bounded by $\Gamma_g + \sigma_g^2$ by (A2)–(A3), and the smoothness term is absorbed by the step-size condition. The restriction to a block is valid because the gradient of block g is bounded blockwise by the same assumptions. □

The descent-lemma step of the cited theorem bounds $\mathbb{E}\|\nabla F(\bar{\theta}_r)\|^2$ by the progress term, the variance term and L^2 times the *total* deviation $\frac{1}{N}\sum_k \|\theta_k - \bar{\theta}\|^2$. Because the blocks are disjoint, the total deviation is the sum over g of the blockwise deviations, each bounded by the lemma with its own K_g; summing and averaging over rounds gives Proposition 1 of the main text with constants inherited from the cited theorem. We make no claim about the constants beyond their existence. Setting $K_g = 1$ for a single block returns the cited bound with τ local steps.

What the controller sees. By the same lemma the dispersion of the received slices of tier g after K_g rounds is of order $\eta^2 K_g^2 \tau^2 (\Gamma_g + \sigma_g^2)$; normalising by $\sum_j \|\bar{\theta}_j\|^2$ gives the scale-free $\hat{\Gamma}_g$ of Eq. (7) of the main text (the dispersion estimator). The controller therefore minimises $\sum_g K_g^2 \hat{\Gamma}_g$ where $\hat{\Gamma}_g$ already carries the factor K_g^2 of the interval that produced it; it is a feedback loop on a proxy, and its stability is observed (Fig. 4 of the main text), not proved.

III. S3. Differential-privacy accounting

Per-client clipping to C makes the ℓ_2 sensitivity of the sum to the addition or removal of one client exactly C, and the fixed denominator qN does not depend on the realised cohort. With Poisson sampling at rate q each release is the subsampled Gaussian mechanism, whose RDP at integer order α is $\frac{1}{\alpha-1}\log\sum_{k=0}^{\alpha}\binom{\alpha}{k}(1-q)^{\alpha-k}q^k \exp\frac{k^2-k}{2\sigma^2}$ [Mironov, Talwar and Zhang 2019, Sec. 3.3]; for $q = 1$ it is $\alpha/(2\sigma^2)$ [Mironov 2017, Prop. 7]. Composition over the R_{tx} planned releases is additive, and (ε, δ)-DP follows from [Mironov 2017, Prop. 3], minimised over $\alpha \in \{2, \dots, 256\}$ (and a fine fractional grid for $q = 1$). The accountant is valid only if the set of release rounds is independent of the private data, which is why only FedAvg, FedProx, FedPer, the fixed period and TierFed-DP are run under DP in the main text. The sum is computed in log space; the implementation matches the TensorFlow-Privacy MNIST reference ($\varepsilon \approx 3.0$ at $q = 256/60000$, $\sigma = 1.1$, 60

TABLE I

SUPERSEDED EQUAL-σ COMPARISON OF V3 (PER-CLIENT CLIPPING $C = 1$, 10 SEEDS, $q = 1$). THE BUDGETS OF THE ADAPTIVE SCHEDULERS (TIERFED, ADACOMM) DO NOT ACCOUNT FOR THEIR DATA-DEPENDENT SCHEDULES AND ARE THEREFORE NOT VALID DP GUARANTEES: $(\varepsilon, \delta = 10^{-5})$ FROM THE EXACT RÉNYI ACCOUNTANT, NUMBER OF CHARGED (TRANSMITTING) ROUNDS, FINAL AUROC AND AUPRC. THE SAME MECHANISM IS APPLIED TO EVERY METHOD.

Method / σ	n	ε	Charged rounds	AUROC	AUPRC
AdaComm / 0.5	10	43.2	8	0.6223 ± 0.0083	0.158
AdaComm / 1.0	10	17.6	8	0.6076 ± 0.0115	0.148
AdaComm / 2.0	10	7.8	8	0.5469 ± 0.0135	0.123
AdaComm / 4.0	10	3.6	8	0.5460 ± 0.0159	0.108
FedAvg / 0.5	10	194.4	60	0.5990 ± 0.0156	0.147
FedAvg / 1.0	10	67.2	60	0.5008 ± 0.0096	0.096
FedAvg / 2.0	10	26.1	60	0.5143 ± 0.0227	0.099
FedAvg / 4.0	10	11.2	60	0.5150 ± 0.0151	0.094
FedPer / 0.5	10	194.4	60	0.6250 ± 0.0081	0.161
FedPer / 1.0	10	67.2	60	0.5669 ± 0.0136	0.133
FedPer / 2.0	10	26.1	60	0.5196 ± 0.0151	0.111
FedPer / 4.0	10	11.2	60	0.5198 ± 0.0161	0.105
Fixed schedule / 0.5	10	132.4	37	0.6248 ± 0.0107	0.159
Fixed schedule / 1.0	10	47.7	37	0.5923 ± 0.0163	0.142
Fixed schedule / 2.0	10	19.2	37	0.5372 ± 0.0185	0.118
Fixed schedule / 4.0	10	8.5	37	0.5155 ± 0.0167	0.109
TierFed / 0.5	10	76.8	18	0.6230 ± 0.0100	0.160
TierFed / 1.0	10	29.4	18	0.6100 ± 0.0095	0.151
TierFed / 2.0	10	12.4	18	0.5644 ± 0.0114	0.131
TierFed / 4.0	10	5.7	18	0.5282 ± 0.0094	0.114

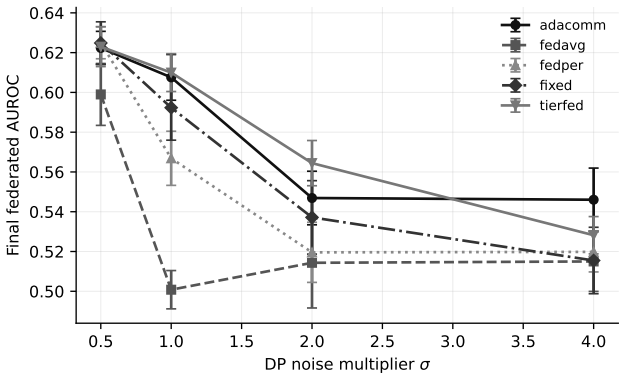


Fig. 1. Utility against the DP noise multiplier for every method under the identical mechanism (main text, Eq. (8)).

epochs) to within 0.05. Model selection on held-out data is not covered by the budget, so DP results are final-round values.

The superseded equal-σ comparison. Version 3 of this article compared every method at equal noise multiplier with full participation. It is reproduced below because it was circulated, with the caveat that the budgets reported for adaptive TierFed and AdaComm are not valid: both choose when to transmit from client updates or losses, and that choice is not charged by the accountant.

Table I and Fig. 1 were presented in v3 as the clearest effect of the study. Under client-level DP, every transmitting round is a release that the accountant charges for and into which noise is injected, so the number of transmitting rounds sets both the budget and the cumulative noise. At $\sigma = 1$, full-rate FedAvg collapses to 0.5008 and FedPer reaches 0.5669, both at $\varepsilon = 67.2$; TierFed reaches 0.6100 at $\varepsilon = 29.4$ (TierFed minus FedPer +0.043, $p =< 10^{-4}$). At $\sigma = 2$ the margin over FedPer is +0.045 ($p =< 10^{-4}$) at less than half the budget. Against AdaComm, the strongest round-sparse baseline, the comparison is mixed and we report it as such: level at $\sigma = 0.5$ (+0.001, $p = 0.604$) and $\sigma = 1$ (+0.002, $p = 0.443$), better at $\sigma = 2$ (+0.018, $p =< 10^{-4}$), and worse at $\sigma = 4$ (-0.018, $p = 0.0039$), where AdaComm's eight releases incur less noise than TierFed's eighteen. The operational reading is that, under DP, the number of rounds a site transmits is the dominant design variable, and a communication quota is also a privacy quota. None of the budgets reported is claimed to constitute meaningful privacy; they are reported as computed so that the trade-off is measured correctly.

IV. S4. PER-SEED TABLES

Every run's final metrics are in `results/tables/collected_all_raw.csv` in the released repository. Total runs at build time: 1473.

V. S5. SGD ARM

The SGD arm (learning rate 5×10^{-2}) is reported in Table VI of the main text (baseline sweeps).

VI. S6. EXCLUDED SITES ON THE IMAGING BENCHMARK

Number of sites whose patient-grouped held-out split contained a single class (identical for every method, per seed): between 3 and 9 of 20 depending on the seed (read from `n_clients_single_class_final` in the logs).